Memory Transfer Learning: How Memories are Transferred Across Domains in Coding Agents

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HarnessRSI 67 ★★
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Paper

Code

1 What it reuses, what is shared

1 Source runs
Agent attempts other benchmarks (all but the test one); an LLM judge labels each attempt

2 4 formats, one per attempt (our reading)
Trajectory · Workflow · Summary · Insight
raw commands ——— abstract, no file names

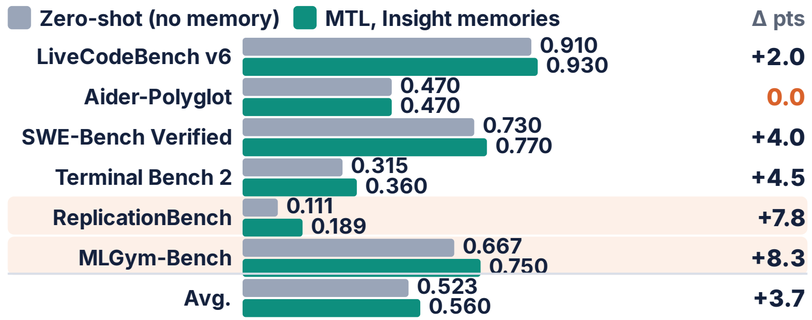
3 Pool for a test benchmark
all benchmarks but the test one; per model and format; no filter described by the paper

4 Retrieve 3 memories
a model-written 4-5 sentence plan is the query (Trajectory: task text); top 3 by similarity

5 Solve the target task
3 memories go into the system prompt; mini-swe-agent; score = Pass@3 (three runs)

no weights change, no fine-tuning
cost of building the pool: not reported

2 Headline: +3.7 points on average, uneven



GPT-5-mini, Pass@3, Table 1; shaded: two largest gains. Avg is task-pooled (our reading). No SD or CI. DeepSeek V3.2 +2.6, Qwen3-Coder +1.8, Insight only. Insight is the best of 4 formats on these same test tasks; no dev split.

4 Where the gain is absent or thin

- **Aider-Polyglot 0.0** for GPT-5-mini; DeepSeek −1.0 there; Qwen3-Coder 0.0 on ReplicationBench, MLGym
- **MLGym +8.3 is 1 task** of ~12‡; ReplicationBench +7.8 is ~7‡ of the 18‡ net tasks (our task-count reading)
- **Pass@1 gain is +1.9** (0.435 → 0.454), about half of Pass@3 +3.7
- Trajectory is below zero-shot on Terminal Bench 2 and MLGym; "all MTL variants outperform the zero-shot baseline" is average-only
- Full-pool gain +2.8‡ (Table 6: 0.543 − 0.515) is ReplicationBench-driven; "9 domains" and the 431-memory pool unexplained

3 Controls the paper reports

Memory format · avg Pass@3, GPT-5-mini (Table 1)



Memory baselines (Table 2; model not named)

Pass@3	#mem	LCB	SWEB	RepliB	Avg
Zero-shot	-	.910	.730	.111	.584
ReasoningBank	97	.920	.750	.133	.601
AgentKB	5,899	.920	**.720**	**.200**	.613
MTL	431	.930	**.770**	.189	**.630**

MTL row = Table 1 Insight cells (our reading); ReasoningBank: in-domain memories, AgentKB: out-of-domain (general reasoning).

Abstraction (Table 4) · Retrieval (Table 7)
Task-specific 0.523 vs task-agnostic 0.534 insights; no variance, metric not named in table. LLM reranking .598 and rewriting .608 (Table 7, Pass@3) do not beat plain embedding .630.

Memories from another model (Table 6, Pass@1)

Pass@1 · 3 benchmarks	zero-shot	other models	own
target GPT-5-mini	.515	DS .518 · Qwen .528	.543
target DeepSeek V3.2	.486	GPT .501	.511
target Qwen3-Coder	.402	GPT .413	**.413**

DS = DeepSeek V3.2; its memories ≈ zero-shot for GPT (.518 vs .515); Qwen gain is LiveCodeBench only; .413 tie at printed precision.

Takeaway

Memories built on other benchmarks give a small, uneven average gain (none on Aider-Polyglot, largest on ReplicationBench and MLGym); Insight-style abstract memories do best on average. The paper reports no variance or cost, so we read the direction as plausible and the size as uncertain.

Our read · HarnessRSI

Score 67 · ★★ · evidence level: retained

Reports: pool without the test benchmark, 3 models, 4 formats, ablation, 2 memory baselines. **Does not report:** SD / CI, cost, leakage check, judge accuracy, random-memory control.